



PSG Institute of Technology and Applied Research, Coimbatore

RSCPF_ML: Region-Specific Crop Price Forecasting through Machine Learning Techniques

Team Members

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Project Guide

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Outline of the Presentation

- Introduction and Problem Statement
- Literature Review and Gaps Identified
- Proposed Methodology for Region-Specific Crop Price Forecasting using ML Algorithms
- Hardware and Software Specifications
- Modules, Roles and Responsibilities
- Snapshots and Results
- Conclusions and Future Enhancements
- References

Integration of Environmental and Regional Factors

- **Impact of climatic variability**
 - Weather conditions influence crop yield and pricing
- **Importance of region-specific analysis**
 - Local variations affect market trends
- **Multi-factor influence on price behavior**
 - Seasonality, supply, and environment interact
- **Growing demand for adaptive forecasting systems**
 - Static models became insufficient

Need for an Advanced Forecasting Framework

- **Persistent agricultural price volatility**
 - Influenced by multiple dynamic factors
- **Structural gaps in existing approaches**
 - Inability to capture multi-dimensional interactions
- **Requirement for integrated data analysis**
 - Combining historical and environmental insights
- **Need for reliable and risk-aware forecasting**
 - Supporting informed market participation

Problem Statement

Crop price fluctuations make accurate and timely forecasting difficult for farmers and traders.

- Crop prices fluctuate significantly due to market volatility, climate change, droughts, floods, and other environmental factors, making accurate forecasting difficult.
- Traditional forecasting models rely mainly on historical price data and fail to integrate real-time weather and market information.
- Advanced machine learning models provide high prediction accuracy but often lack transparency and interpretability for end users.
- There is a need for a region-specific, accurate, and explainable crop price forecasting system to support better decision-making for farmers, traders, and policymakers.

Literature Review

Machine Learning Based Agricultural Price Forecasting for Major Food Crops in India Using Environmental and Economic Factors (Krishna, P. A., Narayana, G. V. S., Kotha, S. K., & Pattnayak, D. (2025).	• Studies price changes in major food crops in India that make it hard for farmers and markets to plan • Uses machine learning models that combine weather data and economic factors to predict crop prices • Shows better and more stable price predictions than traditional forecasting method • Gives limited explanation of the algorithms, depends heavily on data quality, and is hard to reproduce
Predicting Prices of Staple Crops Using Machine Learning (Theofilou, A., Nastis, S. A., Michailidis, A., Bournaris, T., & Mattas, K. (2025). Predicting Prices of Staple Crops Using Machine Learning: A Systematic Review of Studies on Wheat, Corn, and Rice. Sustainability, 17(12), 5456. https://doi.org/10.3390/su17125456)	• Reviews earlier studies to understand how machine learning is used to predict staple crop prices • Analyzes research using a PRISMA-based systematic review of ML, deep learning, and time-series models • Finds that hybrid deep-learning models usually perform better than older prediction methods • Explains research trends, strengths, and gaps in existing studies • Does not develop a new prediction model and considers only Scopus-indexed papers

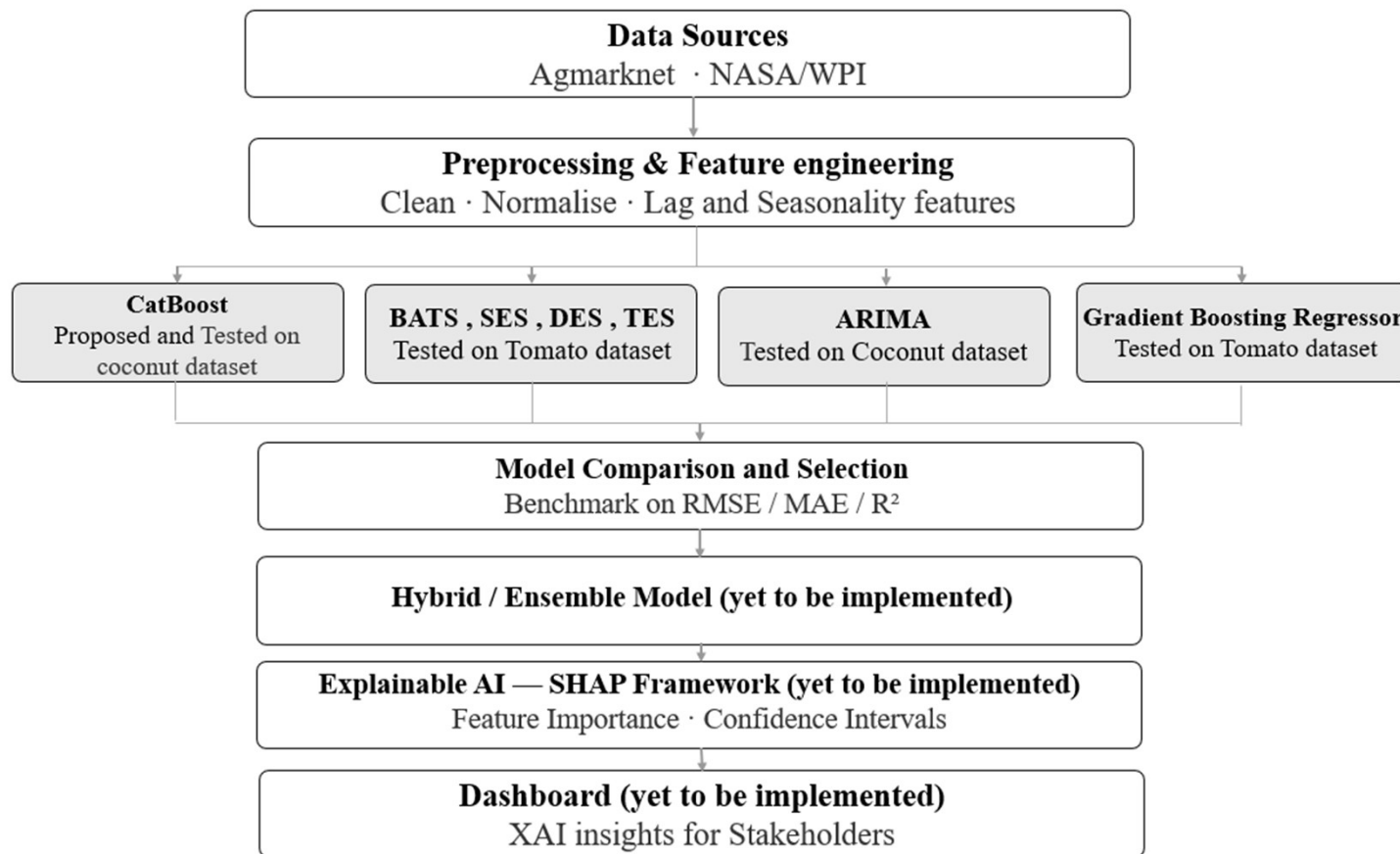
Enhancing agricultural commodity price forecasting with deep learning Manogna, R.L., Dharmaji, V. & Sarang, S. Enhancing agricultural commodity price forecasting with deep learning. <i>Sci Rep</i> 15, 20903 (2025). https://doi.org/10.1038/s41598-025-05103-z	• Uses daily wholesale price data for 23 agricultural commodities from January 2010 to June 2024 to analyse forecasting models. • Compares traditional statistical models (e.g., ARIMA) with machine learning (SVR, XGBoost) and deep learning (MLP, RNN, LSTM, GRU, ESN) for price forecasting. • Evaluates model performance using standard error metrics such as RMSE, MAE, MAPE, and RNMSE. • Finds that deep learning models, especially LSTM and GRU, consistently outperform traditional and classical machine learning methods.
Avinash, G., G. H. Harish Nayak, Moumita Baishya, B. Samuel Naik, and Karthik V.C. 2024. “Enhancing Agricultural Commodity Price Forecasting Using Generative Models: A Deep Learning Approach”. <i>Journal of Scientific Research and Reports</i> 30 (10):1-11. https://doi.org/10.9734/jsrr/2024/v30i102424.	• Uses daily soya commodity price data (2010–2020) and confirms the data is non-stationary and highly nonlinear using ADF, PP, KPSS, and BDS tests • Proposes a GAN-based deep learning model, where the generator uses GRU layers and the discriminator uses CNN layers • Compares GRU, LSTM, Bi-LSTM, and GAN models using RMSE as the main performance metric • Finds that the GAN model performs best, achieving the lowest RMSE (251.60) compared to Bi-LSTM (388.04) and LSTM (449.72)

Crop Price Prediction and Forecasting System using Supervised Machine Learning Algorithms ([1] Rohith R, Vishnu R, Kishore A, Deeban Chakkarawarthi, "Crop Price Prediction and Forecasting System using Supervised Machine Learning Algorithms," International Journal of Advanced Research in Computer and Communication Engineering (IJARCCE), DOI: 10.17148/IJARCCE.2020.9306)	• Addresses uncertainty and losses from crop price fluctuations affecting farmers' income and decision-making. • Develops a crop price prediction and forecasting system using Decision Tree Regression and parameters such as rainfall, Wholesale Price Index (WPI) for forecasting up to 12 months. • Produces detailed price forecasts for the next year, aiming to help farmers increase profits by anticipating price changes. • Limited model complexity and evaluation details, and dependent on rainfall and WPI parameters only, which may not capture all price influencers.
Agriculture Crop Price Prediction Using Machine Learning Bhatakar, A., Tayde, L., Raut, S., Pakhare, A., Bawsakar, T., & Shivaji Chavan. (2025). Agriculture crop price prediction using Machine learning. In <i>International Journal of Interdisciplinary Innovative Research & Development (IJIIRD)</i> (Vol. 09, Issue 02, p. 26).	• Uses historical crop prices (2010–2022) along with weather data and economic indicators like WPI and prediction • Tests Linear Regression, Decision Tree, Random Forest, and SVM, with Random Forest and SVM giving better accuracy • Applies data cleaning, normalization, feature selection, and an 80:20 train–test split • Evaluates performance using MAE, RMSE, and R², showing machine learning improves price prediction reliability • Limited by use of historical data only, no real-time inputs, and no deep learning models implemented

Gaps Identified

- Most existing models rely heavily on historical price data and do not effectively integrate real-time weather, market, and economic information.
- Many studies focus on a single machine learning or deep learning algorithm without performing comprehensive model comparison and optimization.
- Most forecasting models are developed for specific crops or regions, limiting their generalizability and region-specific adaptability.
- Deep learning models achieve high prediction accuracy but often lack interpretability and explainability for farmers and policymakers.
- Many approaches require large, high-quality datasets and are difficult to reproduce or deploy in practical agricultural environments.

Workflow Implementation for Crop Price Prediction



Hardware and Software Specifications

Hardware:

- Processor: AMD Ryzen 7 / Intel Core i7
- Memory: 16 GB RAM
- Storage: 512 GB SSD
- Operating System: Windows 11 (64-bit)
- Graphics: NVIDIA GeForce RTX 3050 Laptop GPU (4 GB VRAM)

Software:

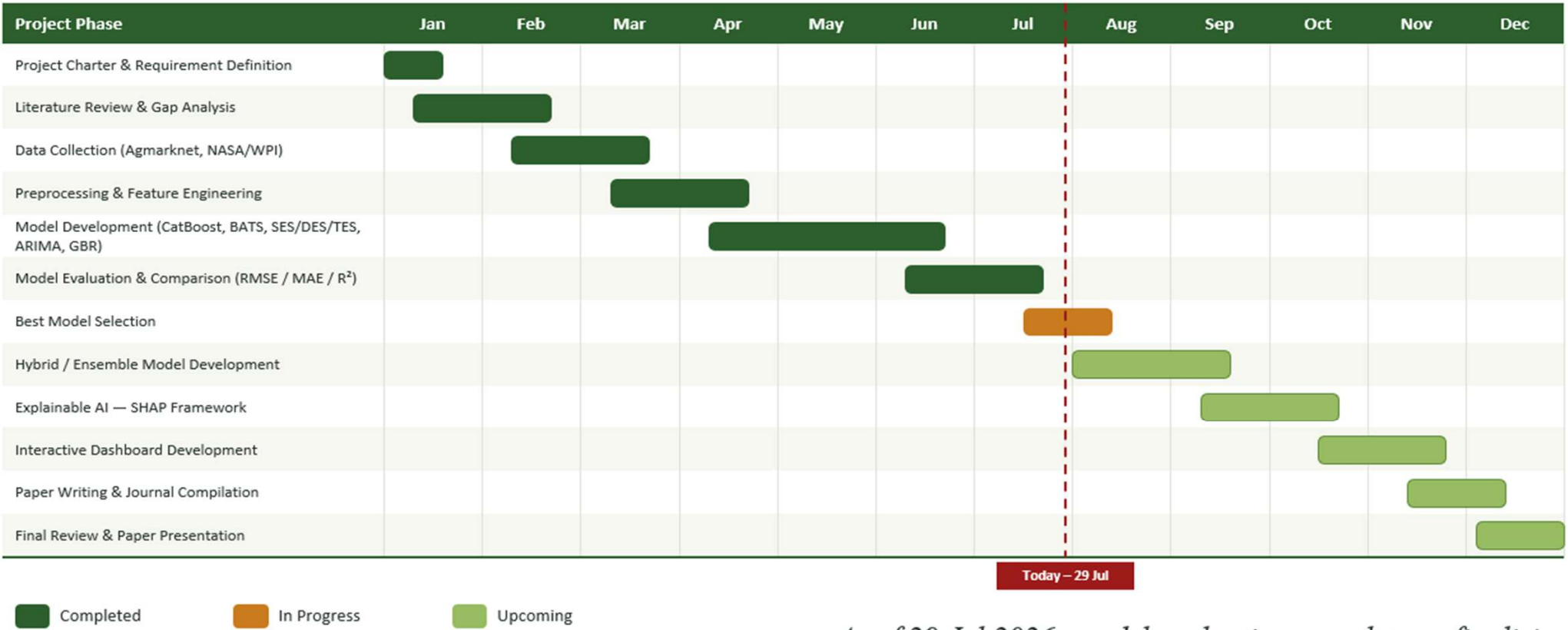
- Python 3.11+
- Google Colab
- Visual Studio Code
- Scikit-learn
- Optuna
- SHAP
- Matplotlib
- Explainable AI

Modules, Roles and Responsibilities

Modules	Description
Data Engineering (Rahul S)	• Collect, clean, and merge crop price and weather datasets. • Perform preprocessing and generate calendar/weather features.
Machine Learning Development (Nithikbalaji S)	• Create lag, rolling, and EMA-based historical features. • Train and optimize ML models
Model Evaluation & Explainability (Rahulkhanna V)	• Evaluate performance using MAE, RMSE, MAPE, and R^2. • Interpret predictions using Feature Importance and SHAP analysis.
Validation & Deployment (Nithika Murugan)	• Validate the model using Timeseries Split and comparative analysis. • Deploy the prediction system and prepare the journal publication.

RSCPF_ML Project Timeline

Region-Specific Crop Price Forecasting through Machine Learning Techniques | January 2026 – December 2026



As of 29 Jul 2026: model evaluation complete — finalizing best-model selection before hybrid/XAI phases.

Model Training Snapshots and Results

```
tscv = TimeSeriesSplit(n_splits=5)

results = []

fold = 1

for train_idx, test_idx in tscv.split(X):

    X_train_cv = X.iloc[train_idx]
    X_test_cv = X.iloc[test_idx]

    y_train_cv = y.iloc[train_idx]
    y_test_cv = y.iloc[test_idx]

    model = CatBoostRegressor(
        **best_params,
        loss_function="RMSE",
        random_seed=42,
        verbose=False
    )

    model.fit(X_train_cv, y_train_cv)
    pred = model.predict(X_test_cv)
    mae = mean_absolute_error(y_test_cv, pred)
    rmse = np.sqrt(mean_squared_error(y_test_cv, pred))
    r2 = r2_score(y_test_cv, pred)
    results.append([fold, mae, rmse, r2])
    fold += 1

cv_results = pd.DataFrame(
    results,
    columns=["Fold", "MAE", "RMSE", "R2"]
)
```

CatBoost for Coconut Dataset

```
from tbats import BATS

bats_model = BATS(
    seasonal_periods=[12]
).fit(train['Price'])

bats_pred = bats_model.forecast(steps=len(test))

plt.figure(figsize=(15,5))
plt.plot(train.index, train['Price'], label='Train')
plt.plot(test.index, test['Price'], label='Actual')
plt.plot(test.index, bats_pred, label='BATS')
plt.legend()
plt.grid(True)
plt.show()

bats_result = evaluate(test['Price'], bats_pred)
print(bats_result)
```

BATS for Tomato dataset

Model Training Snapshots and Results

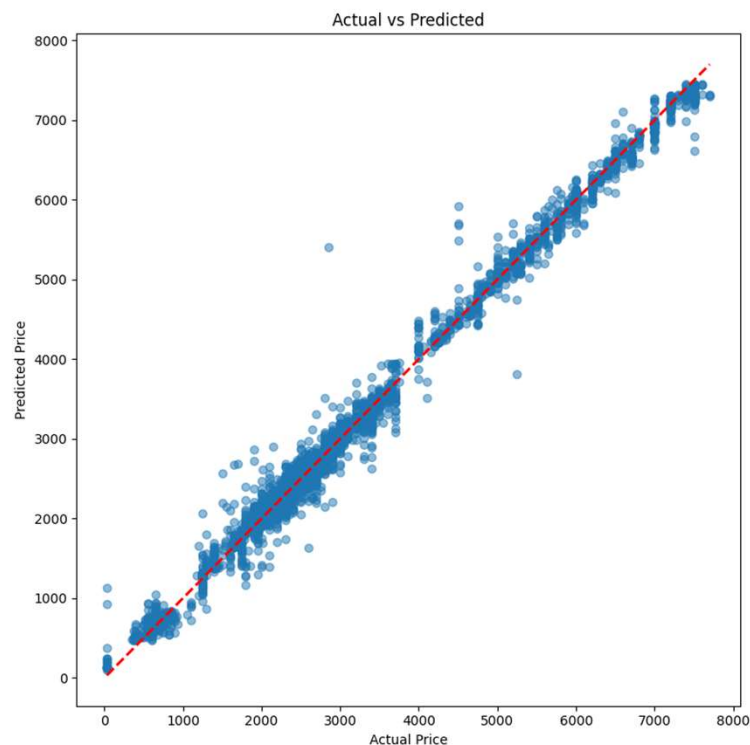
FINAL MODEL PERFORMANCE

Training R² : 0.9928
Testing R² : 0.9950
Train-Test Gap : -0.0022
MAE : 62.88
MSE : 11783.93
RMSE : 108.55
MAPE : 4.57%

Generalization : Excellent

First 20 Predictions

	Actual Price	Predicted Price	Absolute Error
0	1950	2019.895024	69.895024
1	1950	1986.309934	36.309934
2	2050	1991.590934	58.409066
3	2000	2032.611340	32.611340
4	2000	2008.407858	8.407858
5	1950	2011.134195	61.134195
6	2000	1990.651571	9.348429
7	2000	2016.100316	16.100316
8	2000	2011.289825	11.289825
9	2200	2013.788047	186.211953
10	2300	2114.954646	185.045354
11	2300	2216.033956	83.966044
12	2350	2249.476903	100.523097
13	2200	2298.127036	98.127036
14	2100	2196.000912	96.000912
15	2200	2148.593345	51.406655
16	2000	2200.664229	200.664229
17	2100	2121.676575	21.676575
18	2200	2141.662494	58.337506
19	2300	2201.295336	98.704664



FINAL OPTUNA TUNED CATBOOST MODEL

Training Samples : 38411
Testing Samples : 9603
Features Used : 28
Training R² : 0.9928
Testing R² : 0.9950
MAE : 62.88
RMSE : 108.55
MAPE : 4.57%

Results of the CatBoost algorithm after Hyper-Parameter tuning

Model Training Snapshots and Results

	MAE	RMSE	R2	MAPE(%)
Gradient Boosting Regressor	221.4	281.22	0.4512	4.97
Random Forest Regressor	239.27	297.09	0.3875	5.34
XGBoost Regressor	236.79	297.63	0.3853	5.32
Decision Tree Regressor	269.04	345.46	0.1719	6.01

The **Gradient Boosting Regressor** performed best, achieving an **MAE of 221.40**, **RMSE of 281.22**, **R² of 0.4512**, and **MAPE of 4.97%**.

Model	RMSE	MAE	MAPE (%)
SES	1248.40	929.28	29.43
DES	1099.32	857.27	29.02
TES	996.50	854.23	37.41
ARIMA	1273.18	941.23	29.43
SARIMA	1273.18	941.23	29.43
BATS	1560.08	1225.11	39.67
TBATS	1265.22	937.60	29.45

The **TES model** achieved the lowest **RMSE (996.50)** and **MAE (854.23)**, while **DES** recorded the lowest **MAPE (29.02%)** among the evaluated forecasting models.[4]

Conclusions and Future Enhancements

- To Develop an accurate and reliable machine learning prediction framework.
- Improved model performance through feature engineering, Hyperparameter tuning and optimization.
- Enabled data-driven decision-making with explainable predictions.
- Future work includes integrating real-time data and advanced AI models.
- The framework can be extended for scalable real-world deployment and broader applications.

References

1. Kaewchada, S. and Ruang-On, S., *Random forest model for forecasting vegetable prices*. Int. J. Electr. Comput. Eng. (IJECE), 13, 6, 5265–5272, 2023.
2. Singh, N., *Crop Price Prediction Using Machine Learning*. J. Electr. Syst., 10, 2, 2258–2269, 2024.
3. Murugesan, G., *An extrapolative model for price prediction of crops using hybrid ensemble learning techniques*. Int. J. Adv. Technol. Eng. Explor., 10, 4, 1–10, 2023.
4. Vinay, H. T., Basak, P., Ghosh, A., Ojha, S., & Sahu, C. R. (2024). Forecasting of Tomato Price in Karnataka using BATS Model. *Journal of the Indian Society of Agricultural Statistics*, 78(2), 107–113. [[1](#), [2](#)]
5. Meena, K., and B. Chaitra. "A Novel Framework Using Deep Learning Techniques for Ragi Price Prediction in Karnataka." *IEEE Access*, vol. 12, 2024. *IEEE Xplore*, [ieeexplore.org](https://ieeexplore.ieee.org). [[1](#), [2](#), [3](#)]